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I. Use Of Multimeter :

1. ***Measuring Resistance :***

- Turn the multimeter dial to the resistance (Ω) setting.
- Connect the probes to the multimeter : the black probe goes into the COM port, and the red probe goes into the VΩmA port.
- Touch the probes to the two points of the resistor . The display will show the resistance value in ohms.

2. ***Measuring Voltage :***

- Turn the multimeter dial to the voltage (V) setting ,
- Connect the probes ,
- Touch the probes to the two points where you want to measure the voltage. The black probe goes to the ground or negative point, and the red probe goes to the positive point.
- The display will show the voltage across the points.

3 . ***Measuring Current :***

- Turn the multimeter dial to the current (A) setting
- Break the circuit where you want to measure the current.
- Connect the multimeter in series with the circuit by placing the probes at the break point. The black probe goes into the COM port, and

the red probe goes into the appropriate current port (VΩmA for low current, or A for high current).

II. Kirchhoff's Current Law :

1. ***Kirchhoff's Current Law (KCL) :***

- This law states that the total current entering a junction (or node) in an electrical circuit must equal the total current leaving that junction.

sum I_{in} = sum I_{out}

2. ***Kirchhoff's Voltage Law (KVL) :***

- This law states that the total sum of the electric potential differences (voltages) around any closed loop in a circuit must equal zero.
- Mathematically, it can be expressed as:

SUM V = 0

III. DIODE :

A diode is an semiconductor device that allow electricity to flow in one direction and block it from the opposite direction .

In order for a semiconductor material to be useful , it needs to be doped by an impurity atom that could be a pentavalent atom to produce a N-type material or a trivalent atom to produce a P-type . When we take a small piece of semiconductor and doped the half with a pentavalent atom and the other half with a trivalent, it will result in a semiconductor device with p-type and n-type portions with a PN junction between them which is basis of diode . The free electrons from the n-type part will diffuse to fill up the holes in p-type part ; this diffusion will create an electric field across the junction that blocks the free electrons from the n region to flow across the junction .

A diode a two expected output character :

- ***Forward bias mode*** : the diode conduct electricity (positive terminal of the battery on the P-type and negative terminal on the N-type)

- ***Reverse bias mode*** : no current will flow (positive terminal of the battery on the N-type and negative terminal on the P-type)

IV. TYPE OF DIODES :

- **PN junction Diode :**
  normal or standard type of diode , used in radio frequency or other low current applications.

- **Zener Diode :**
  used to provide a stable reference voltage (breaks down when a particular voltage is reached) .

- **Gunn Diode :**
  used for producing microwave signal .

- **Laser Diode :**
  used in DVDs, CD drives , laser light pointers .

- **Schottky Diode**
- **Light emitting Diode**
- **Photodiode**
- **PIN Diode**
- **Step recovery Diode**